

Hongyi Mei

Research Interests

Focus Astrostatistics, AI4Science, AI Safety/Responsibility, Social Justice

Education

Sep 2024–Present **B.Sc. Astronomy Specialist, Statistics Major, Computer Science Minor**, *University of Toronto, St George*, Toronto, Canada

○ **Astronomy & Statistics coursework: 3.8/4.0; Dean's List Scholar.**

May–Aug 2026 **Summer Research Exchange Program Student (SREP)**, *ETH Zürich*, Zürich, Switzerland
Institute for Particle Physics and Astrophysics. SREP + IE Award (4,500 CAD total).

Manuscripts

In prep. A. Salmi*, **H. Mei***, C. Dorn, *Can Super-Earths Be Born Rocky? Separating Survey Selection from Formation in the Deficit of Large Rocky Planets at Low Insolation*, in preparation (2026). *Equal contribution.

Research Experience

May 2026–Present **Survey Selection and the Origin of Cold Rocky Super-Earths**, *ETH Zürich*, Zürich, Switzerland, Supervisors: Prof. Caroline Dorn, Annika Salmi
Python, NumPy, SciPy, pandas, AstroPy, Matplotlib, P-Pop. **Role:** designed the code and methods, designed all of the analysis, and made every figure in the paper.

- Built detection models for *Kepler*, *TESS* and precision radial velocity (HARPS/NIRPS), and tested each against its official pipeline on 7,995 KOIs, 7,307 TOIs and 2,510 published RV semi-amplitudes. Calibrated models to match official values to 0.93–1.00.
- Designed a prior-free population generator for simulating planets around volume-limited 60 pc *Gaia* stellar samples. It replaces a ~6-hour P-pop run (Kammerer & Quanz, 2018) with under 1 s per 10^6 planets at the same detectability, so selection can be measured without assuming an occurrence rate.
- Turned two formation scenarios into priors on the mass–radius plane and ran them through the detection model, then compared posterior-predictive volatile fractions against the NASA Exoplanet Archive across 5,000 Monte Carlo draws.

May–Aug 2025 **Data Science Research Assistant (ROP299)**, *Munk School of Global Affairs*, Toronto, ON, Supervisor: Prof. Lynette Ong

- Manually coded **500+** protest cases into a team spreadsheet of **7,000+** categorical records from five news/NGO sources.
- Analyzed monthly trends and provincial distribution from 2023–2026, focusing on housing protests, migrant-labor cases; found over a quarter of the cases from Guangdong and Henan.

Jan–Feb 2025 **Statistical Analyst (Research Project)**, *University of Toronto*, Toronto, ON, Supervisors: Prof. Scott Schwartz & Dr. Kiffer Card

- Built OLS and logistic-regression models predicting PHQ-9 depression scores with various socialize activity (focusing on in-person versus online). In-person interaction outperformed digital connection for well-being ($p < 0.001$).

Technical Skills

Science Physics simulation, Bayesian analysis, Monte Carlo estimation, survey selection and completeness correction, regression, model calibration, neural networks.

Languages Python, Java, C++, $\text{\LaTeX}$

Libraries NumPy, SciPy, pandas, AstroPy, PyTorch

Selected Projects

- Nov 2025 **Passion Finder — AI-Powered UofT Course Explorer**, *Java*
- Built an LLM-powered course recommender for a course final project. Wrote **35%** of the codebase in Clean Architecture and the API integration in a 5-person team.
- Sep 2025 **Neural Networks from Primitives**, *PyTorch*
- Implemented and trained a CNN (MNIST), a recurrent network and a multilayer perceptron from PyTorch primitives instead of prebuilt model libraries. Covered backpropagation, optimization and regularization end to end.
- Sep 2025 **Reinforcement Learning Playground**, *Python*
- Implemented Monte Carlo state-value estimation for a stochastic grid-world (FrozenLake), comparing first-visit value estimates against episode count.
- Mar 2023 **Canadian Young Physicists' Tournament — Oscillating Sphere**, *Team Captain, Silver Medal*
- Captained a team of 5; built a Helmholtz coil pair and the full apparatus myself, fitting results to theory within 6.5%.

Awards & Honours

- 2026 Summer Research Exchange Program Scholarship (3,000 CAD)
- 2026 IE Award (1,500 CAD)
- 2025 Dean's List Scholar, University of Toronto
- 2025 In-Course Scholarship (500 CAD), New College, University of Toronto

Community

- 2026–Present Volunteer and guide, AstroTours public outreach, David A. Dunlap Department of Astronomy & Astrophysics.
- 2023–Present Founding member, grandparents diary project — wrote poetry with seniors at Yee Hong Centre and Amica Bronte Harbour; featured in InsideHalton and The Globe and Mail.
- 2022–Present Assistant Instructor, Kumon Canada — mentored 100+ K–12 students in mathematics and reading.

References

- Prof. Caroline Dorn Assistant Professor for Exoplanets, Institute for Particle Physics and Astrophysics, ETH Zürich. Research supervisor, May 2026–Present.
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